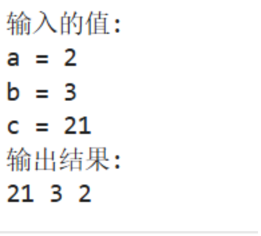
1. Flowchart

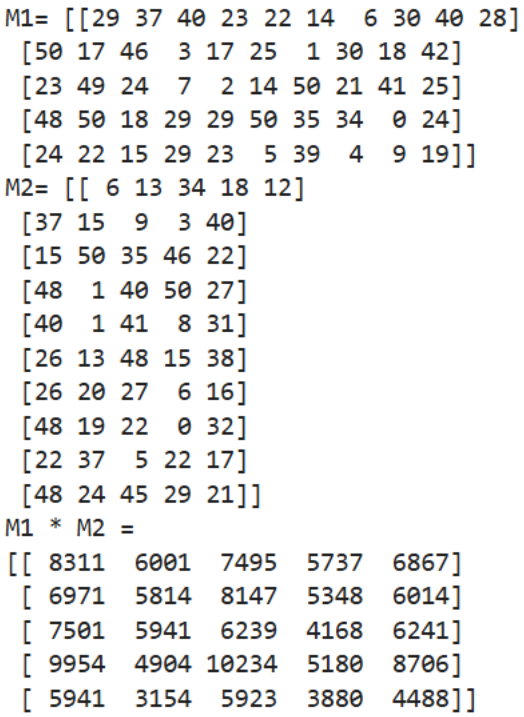
根据流程图编写函数，比较三个数大小，按照指定顺序输出结果。



```
输入的值:
a = 2
b = 3
c = 21
输出结果:
21 3 2
```

2. Matrix multiplication

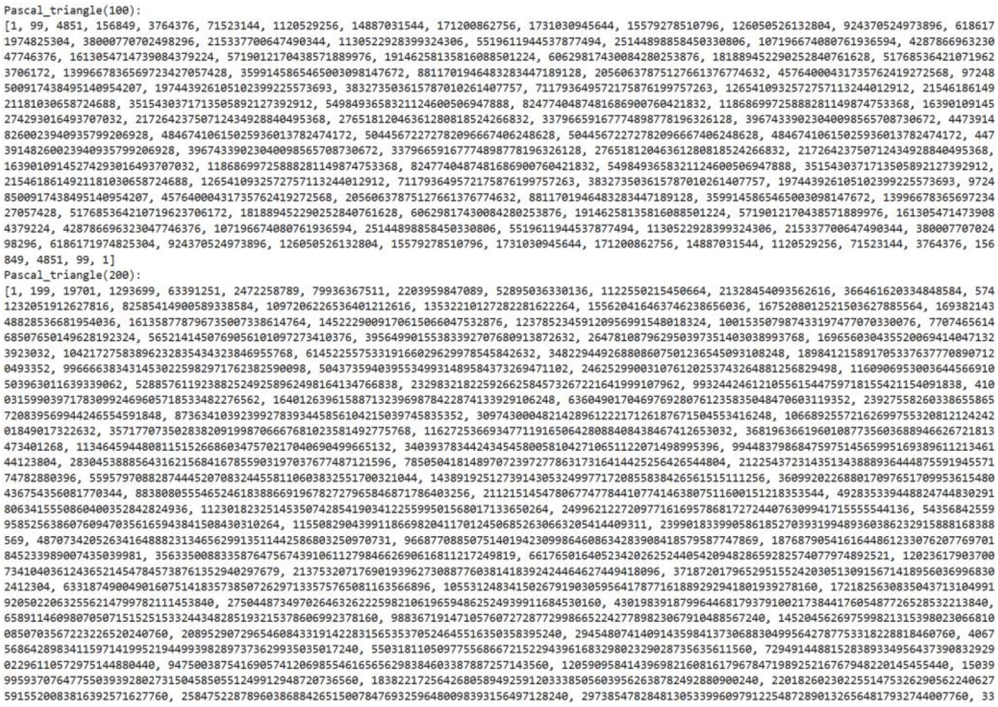
随机生成两个矩阵，通过 for 循环实现矩阵乘法，并输出计算结果。



```
M1= [[29 37 40 23 22 14  6 30 40 28]
 [50 17 46  3 17 25  1 30 18 42]
 [23 49 24  7  2 14 50 21 41 25]
 [48 50 18 29 29 50 35 34  0 24]
 [24 22 15 29 23  5 39  4  9 19]]
M2= [[ 6 13 34 18 12]
 [37 15  9  3 40]
 [15 50 35 46 22]
 [48  1 40 50 27]
 [40  1 41  8 31]
 [26 13 48 15 38]
 [26 20 27  6 16]
 [48 19 22  0 32]
 [22 37  5 22 17]
 [48 24 45 29 21]]
M1 * M2 =
[[ 8311  6001  7495  5737  6867]
 [ 6971  5814  8147  5348  6014]
 [ 7501  5941  6239  4168  6241]
 [ 9954  4904 10234  5180  8706]
 [ 5941  3154  5923  3880  4488]]
```

3. Pascal triangle

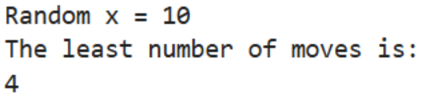
编写函数计算杨辉三角的指定行



```
Pascal_triangle(100):
[1, 99, 4851, 156849, 3764376, 71523144, 1120529256, 14887031544, 171200862756, 1731030945644, 15579278510796, 126050526132804, 924370524973896, 618617
1974825304, 38000770702498296, 215337700647490344, 1130522928399324306, 5519611944537877494, 25144898858450330806, 107196674080761936594, 4287866963230
47746376, 1613054714739084379224, 5719012170438571889976, 19146258135816088501224, 60629817430084280253876, 181889452290252840761628, 51768536421071962
3706172, 1399667836569723427057428, 3599145865465003098147672, 8811701946483283447189128, 20560637875127661376774632, 45764000431735762419272568, 97248
500917438495140954207, 197443926105102399225573693, 383273503615787010261407757, 711793649572175876199757263, 1265410932572757113244012912, 21546186149
21181030658724688, 3515430371713505892127392912, 5498493658321124600506947888, 8247740487481686900760421832, 11868699725888281149874753368, 16390109145
274293016493707032, 21726423750712434928840495368, 27651812046361280818524266832, 33796659167774898778196326128, 39674339023040098565708730672, 4473914
8260023940935799206928, 48467410615025936013782474172, 50445672272782096667406248628, 50445672272782096667406248628, 48467410615025936013782474172, 447
39148260023940935799206928, 39674339023040098565708730672, 33796659167774898778196326128, 27651812046361280818524266832, 21726423750712434928840495368,
16390109145274293016493707032, 11868699725888281149874753368, 8247740487481686900760421832, 5498493658321124600506947888, 3515430371713505892127392912,
2154618614921181030658724688, 1265410932572757113244012912, 711793649572175876199757263, 383273503615787010261407757, 197443926105102399225573693, 9724
8500917438495140954207, 45764000431735762419272568, 20560637875127661376774632, 8811701946483283447189128, 3599145865465003098147672, 13996678365697234
27057428, 517685364210719623706172, 181889452290252840761628, 60629817430084280253876, 19146258135816088501224, 5719012170438571889976, 161305471473908
4379224, 42878669632304774 6376, 107196674080761936594, 25144898858450330806, 5519611944537877494, 1130522928399324306, 215337700647490344, 380007707024
98296, 6186171974825304, 924370524973896, 126050526132804, 15579278510796, 1731030945644, 171200862756, 14887031544, 1120529256, 71523144, 3764376, 156
849, 4851, 99, 1]
Pascal_triangle(200):
[1, 199, 19701, 1293699, 63391251, 2472258789, 79936367511, 2203959847089, 52895036330136, 1122550215450664, 21328454093562616, 366461620334848584, 574
1232051912627816, 82585414900589338584, 1097206226536401212616, 13532210127282281622264, 155620416463746238656036, 1675208012521503627885564, 169382143
48828536681954036, 161358778796735007338614764, 1452229009170615066047532876, 12378523459120956991548018324, 100153507987433197477070330076, 7707465614
68507650149628192324, 5652141450769056101097273410376, 39564990155383392707680913872632, 264781087962950397351403038993768, 169656030435520069414047132
3923032, 10421727583896232835434323846955768, 61452255753319166029629978545842632, 348229449268808607501236545093108248, 18984121589170533763777089071 2
0493352, 9966663834314530225982971762382590098, 50437359403955349931489584373269471102, 246252990031076120253743264881256829498, 1160906953003644566910
503963011639339062, 5288576119238825249258962498164134766838, 23298321822592662584573267221641999107962, 99324424612105561544759718155421154091838, 410
031599039717830992469605718533482276562, 1640126396158871323969878422874133929106248, 6360490170469769280761235835048470603119352, 2392755826033865 5865
7208395699442465545918 48, 87363410392399278393445856104215039745835352, 309743000482142896122217126187671504553416248, 10668925572162699755320812124242
01849017322632, 3571770735028382091998706667681023581492775768, 11627253669347711916506428088408438467412653032, 36819636619601087735603688946626721813
473401268, 113464594480811515266860347570217040690499665132, 340393783442434545800581042710651122071498995396, 9944837986847597514565995169389611213461
44123804, 2830453888564316215684167855903197037677487121596, 7850504181489707239727786317316414425256426544804, 212254372314351343888936444875591945571
74782880396, 55957970882874445207083244558110603832551700321044, 143891925127391430532499771720855838426561515111256, 360992022688017097651709953615480
43675435608177034 4, 883808055546524618388669196782727965846871786403256, 2112151454780677477844107741463807511600151218353544, 492835339448824744830291
806341555086040035284282 4936, 11230182325145350742854190341225599501568017133650264, 24996212272097716169578681727244076309941715555544136, 54356842559
958525638607609470356165943841508430310264, 115508290439911866982041170124506852630663205414409311, 239901833990586185270393199489360386232915888168388
569, 48707342052634164888231346562991351144258680325097073 1, 9668770885075140194230998646086342839084185795877478 69, 187687905416164486123307620776970 1
84523398900743503998 1, 35633500883358764756743910611279846626906168112172498 19, 66176501640523420262524405420948286592825740779748925 21, 12023617903700
7341040361243652145478457387613529402976 79, 2137532071769019396273088776038141839242446462744941809 6, 3718720179652951552420305130915671418956036996830
2412304, 63318749004901607514183573850726297133575765081163566896, 1055312483415026791903059564178771618892929418019392781 60, 172182563083504371310499 1
92050220632556214799782111453840, 275044873497026463262225982106196594862524939911684530160, 430198391879964468179379100217384417605487726528532213840,
658911460980705071515251533244348285193215378606992378160, 988367191471057607272877299866522427789823067910488567240, 14520456269759982131539802306681 0
085070356722322652024076 0, 2089529072965460843319142283156535370524645516350358395240, 294548074140914359841373068830499564278775331822881846076 0, 4067
56864289834115971419952194499398289737362993503501724 0, 5503181105097755686672152294396168329802329028735635611560, 72949144881528389334956437390832929
022961105729751448804 40, 947500387541690574120698554616565629838460338788725714356 0, 12059095841439698216081617967847198925216767948220145455440, 15039
99593707647755039392802731504585055124991294872073656 0, 18382217256426805894925912033385056039562638782492880900240, 22018260230225514753262905622406 27
59155200838163925716277 60, 258475228789603868842651500784769325964800983931564971282 40, 297385478284813053399609791225487289013265648179327440077 60, 33
```

4. Add or double

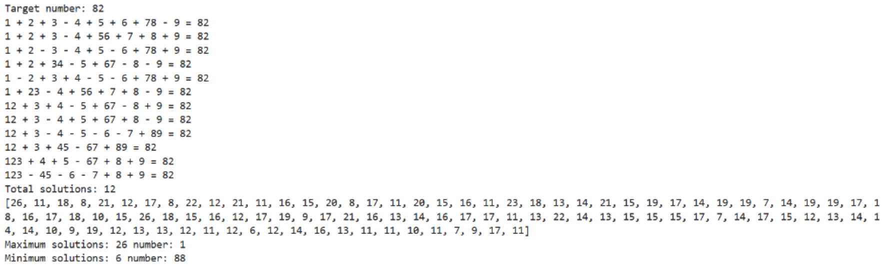
每次加一或者翻倍，计算从 1 到达目标数字所需的最少操作次数。



```
Random x = 10
The least number of moves is:
4
```

5. Dynamic programming

改变数字之间的加减和连接方式，列出所有符合目标值的表达式，并统计解的数量；统计 1-100 解的数量并绘图，找到最大和最小解的数量及对应的数字。



```
Target number: 82
1 + 2 + 3 - 4 + 5 + 6 + 78 - 9 = 82
1 + 2 + 3 - 4 + 56 + 7 + 8 + 9 = 82
1 + 2 - 3 - 4 + 5 - 6 + 78 + 9 = 82
1 + 2 + 34 - 5 + 67 - 8 - 9 = 82
1 - 2 + 3 + 4 - 5 - 6 + 78 + 9 = 82
1 + 23 - 4 + 56 + 7 + 8 - 9 = 82
12 + 3 + 4 - 5 + 67 - 8 + 9 = 82
12 + 3 - 4 + 5 + 67 + 8 - 9 = 82
12 + 3 - 4 - 5 - 6 - 7 + 89 = 82
12 + 3 + 45 - 67 + 89 = 82
123 + 4 + 5 - 67 + 8 + 9 = 82
123 - 45 - 6 - 7 + 8 + 9 = 82
Total solutions: 12
[26, 11, 18, 8, 21, 12, 17, 8, 22, 12, 21, 11, 16, 15, 20, 8, 17, 11, 20, 15, 16, 11, 23, 18, 13, 14, 21, 15, 19, 17, 14, 19, 19, 7, 14, 19, 19, 17, 1
8, 16, 17, 18, 10, 15, 26, 18, 15, 16, 12, 17, 19, 9, 17, 21, 16, 13, 14, 16, 17, 17, 11, 13, 22, 14, 13, 15, 15, 15, 17, 7, 14, 17, 15, 12, 13, 14, 1
4, 14, 10, 9, 19, 12, 13, 13, 12, 11, 12, 6, 12, 14, 16, 13, 11, 11, 10, 11, 7, 9, 17, 11]
Maximum solutions: 26 number: 1
Minimum solutions: 6 number: 88
```